Duration: 110 minutes (Written to take approximately 90 minutes)
Total points available in class: 150, Total points available in take-home: 0 (MCM is awesome, everyone should check it out.)
Instructions: Answer all questions with only the use of your mind, a writing utensil, and a single handwritten 3"x5" note card. Partial credit will be awarded for correct work unless otherwise noted.
You must SHOW ALL WORK to receive full credit.
NO Cell phones. Your phone must be in your bag, and out of reach. If it is out during an exam - your exam is compromised and will be taken by the proctor.

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Problem	Score
1 *(9 pts)*	7
2 *(12 pts)*	12
3 *(5 pts)*	5
4 *(5 pts)*	5
5 *(5 pts)*	4
6 *(8 pts)*	8
7 *(8 pts)*	8
8 *(7 pts)*	5
9 *(7 pts)*	6
10 *(8 pts)*	7
11 *(15 pts)*	10
12 *(15 pts)*	15
13 *(5 pts)*	5
14 *(5 pts)*	5
15 *(5 pts)*	4
16 *(8 pts)*	6
17 *(5 pts)*	3
18 *(6 pts)*	6
19 *(12 pts)*	11
TOTAL*(150 pts)*	132

1. Classify the following partial differential equations: (Order, homogeneity, and linearity) (3pts each)

   (a) $u_{xx} - 2u_y^3 = 0$ 2nd order - homogeneous, non-linear

   (b) $u_x - x^2 u_{tt} = 2$ 2nd order - non-homogeneous, ~~non~~-linear

   (c) $u_{xyxx} - 2u + x^2 = 0$ ~~3rd~~ order - non-homogeneous, linear

2 2 3

2. Show whether the given functions are, in fact, solutions to the given PDEs: (5pts each)

   (a) $u = 4xyz$, $xu_x - 2yu_y + zu_z = 0$

   (b) $u = (x - ct)^3$, $u_t + cu_x = 0$

6

a.) $u = 4xyz$

$u_x = 4yz$

$u_y = 4xz$

$u_z = 4xy$

$x u_x - 2y u_y + z u_z = 0$? (sub in)

$x[4yz] - 2y[4xz] + z[4xy] = 0$?

$4xyz - 8xyz + 4xyz = 0$?

$8xyz - 8xyz = 0$?

$0 = 0$? ✓

u is a solution to the PDE

6

b.) $u = (x-ct)^3$

$u_t = 3(x-ct)^2 \cdot (-c)$

$= -3c(x-ct)^2$

$u_x = 3(x-ct)^2 (1)$

$= 3(x-ct)^2$

$u_t + cu_x = 0$? (sub in)

$[-3c(x-ct)^2] + c[3(x-ct)^2] = 0$?

$-3c(x-ct)^2 + 3c(x-ct)^2 = 0$?

$0 = 0$ ✓ u is a solution to the PDE

6

3. Determine the general solution to the PDE which satisfies the given side condition: $u_y = \cos x$, $u(x, 0) = x^2$ (5pts)

$u_y = \cos(x)$ ($\int$ w.r.t. y)

$\int u_y \, dy = \int \cos(x) \, dy$

$u = y \cdot \cos(x) + g(x)$

$u(x,y) = y \cdot \cos(x) + g(x)$

$u(x,0) = (0) \cdot \cos(x) + g(x)$

$u(x,0) = 0 + g(x)$ (set $g(x) = x^2$)

$g(x) = x^2$ (plug this back in)

✓ $u(x,y) = y \cdot \cos(x) + x^2$

- our final solution is $u(x,y) = y \cdot \cos(x) + x^2$

5

4. Solve the following Convection/Advection problem using method of characteristics: $2u_x - 3u_y = 0$ for its general solution. Pick a function to write as a solution, not just f. (5pts)

$2u_x - 3u_y = 0 \quad (+3u_y)$

$2u_x = 3u_y$

$bx - ay = 0 \rightarrow -3x - 2y = 0$

$u(x,y) = f(-3x-2y)$

$u(x,y) = (-3x-2y)^2$ as a solution.

5 ok

5. What *is* a characteristic curve conceptually? (5pts)

- A characteristic curve is a particular, expanded solution curve to a PDE using the method of characteristics. In short, if we obtained a solution of $u(x,y) = f(x-y)$ then any function of $(x-y)$ would be a solution since it falls on the characteristic curve. yes, but what does that mean further more?

4

6. What conditions determine if a problem is **well-posed**? Explain each condition and what well-posedness tells you about the problem. (8pts)

There are three conditions to determine if a problem is well-posed: Existence, uniqueness, and stability. Existence means that a solution to the problem must exist. Uniqueness means that the solution must be the only solution. Stability means that a small change in the problem, either by its equation or side conditions, will also produce a small change in the solution. Well-posedness implies that the problem can be solved and has a real world solution

8 ok

7. Explain the difference between an initial condition, and a boundary condition. How many of each would you need for the following PDE to be well-posed: $u_{tt} - 2u_x + 4u_{yyy} = 0$? (8pts)

An initial condition is a condition based on time while a boundary condition is based on space. Time is usually represented by t while space is represented by x, y + z. In the above equation, we're given a u_{tt} which means that we'll need two initial conditions to solve the PDE in addition to the boundary conditions. Since the PDE has a u_x and u_{yyy} term, we'll need four total boundary conditions, one for x and three for y, to solve the PDE. In total, we'll need six conditions, two initial and four boundary, to solve the PDE.

8

$L[a \cdot f + b \cdot g] = aL[f] + bL[g]$

8. State the principle of superposition in your own words. When does it hold? (7pts)

There are two ways to determine linearity of an expression and the principle of superposition is one such way. The principle of superposition states that if the linear operator of a constant times a function plus another constant times another function is equal to a constant times the linear operator of the first function plus the product of the second constant & the linear operator of the second function, then linearity is satisfied. This is an extension of the principle itself. not the principle itself.

5

9. Find the general solution for $u_{xx} = 6xe^y$, and identify the homogeneous and particular portions of your solution. (7pts)

$u_{xx} = 6x \cdot e^y$ ($\int$ w.r.t. x)

$\int u_{xx}\,dx = \int 6x \cdot e^y\,dx$

$u_x = 3x^2 \cdot e^y + f(y)$ ($\int$ w.r.t. x)

$\int u_x\,dx = \int 3x^2 \cdot e^y + f(y)\,dx$

$u = x^3 \cdot e^y + x \cdot f(y) + g(y)$ ✓

where $f(y)$ and $g(y)$ are both functions of y.

$u(x,y) = x^3 e^y + x \cdot f(y) + g(y)$

In the above equation, we can let $x=0$ to obtain $u(0,y) = (0)^3 e^y + (0) f(y) + g(y)$

$= g(y)$

This function, $g(y)$, is the homogeneous portion while the particular portion is the ~~$x \cdot f(y)$~~ term.

homogeneous: $xf(y) + g(y)$

particular: $x^3 e^y$

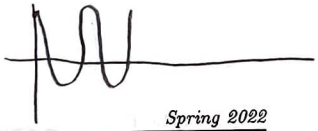
6

10. What is a Fourier Series? And how does it relate to a function $f(x)$? (8pts)

-A function can be represented as an infinite sum of cosine and sine functions. As the sum approaches infinity, the expression becomes a better approximation of $f(x)$, but does never exactly equal the original function. (converges to $f(x)$ though - hard?) This summation is known as a Fourier Series and can be used to approximate values of $f(x)$ as well as the function itself.

7

$F.P = \frac{2\pi}{P}$



11. Determine whether the following functions are periodic (or not); if they are determine their fundamental period: (5pts each)

    (a) $f(x) = 3\cos(4\pi x) + \sin(\pi x)$
    (b) $f(x) = x^2\cos(x)$
    (c) $f(x) = 2 - \cos(2x)$

a.) This function is ~~not~~ periodic 2π

f 2 1

b.) This function is ~~periodic~~ and has a fundamental period of ~~2π~~.

3

c.) This function is periodic and has a fundamental period of

5 $\frac{2\pi}{2} = \pi$.

12. Show whether the following functions are even, odd, or neither: (5pts each)

    (a) $f(x) = x^2\cos(x)$
    (b) $f(x) = 3x^3 - 5x$
    (c) $f(x) = e^x$

a.) Since $\cos(-x) = \cos(x)$, $\cos(x)$ is even. Since $(-x)^2 = x^2$, x^2 is also even. The product of any two even functions is also even. Thus, $x^2\cos(x)$ is even. 5

b.) $\int_{-1}^{1} 3x^3 - 5x\,dx$

$= \frac{3}{4}x^4 - \frac{5}{2}x^2 \Big|_{-1}^{1}$

$= \{[\frac{3}{4} - \frac{5}{2}] - [\frac{3}{4} - \frac{5}{2}]\}$

$= 0$

Since $3(-x)^3 = -3(x)^3$, $3x^3$ is an odd function. Since $-5(-x) = 5(x)$, $-5(x)$ is an odd function. The difference of any two odd functions, is also an odd function. Thus, $3x^3 - 5x$ is an odd function. 5

c.) $f(x) = e^x$

Since $e^{(-x)} \neq -e^x$ and $e^{(-x)} \neq e^x$, e^x is <u>not</u> an even function, nor is it an odd function. Thus e^x is neither.

5

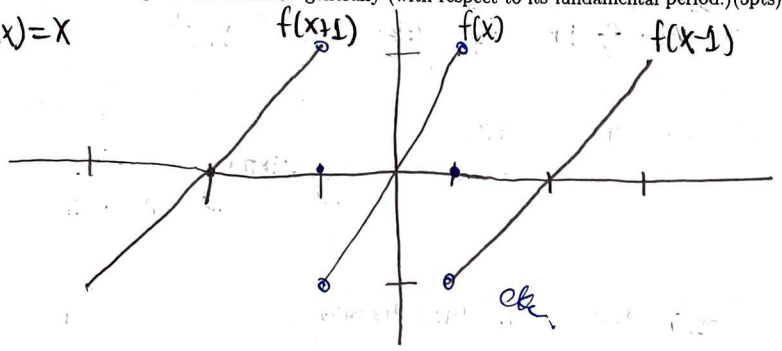
13. What is an inner product? What does it tell you? (5pts)

An inner product a way to determine the orthogonality of two like mathematical enteries. In PDEs, the looks can be represented by the following $\langle f(x), g(x) \rangle = \int_a^b f(x)\, g(x)\, dx$ where f + g are functions.

~5

14. Sketch 3 periods of the Fourier Series for the periodic extension of $f(x) = x$ on $-1 \le x \le 1$ and define the periodic extension generally (with respect to its fundamental period.)(5pts)

Let: $f(x) = x$



$$f(x) = \begin{cases} x & \text{on } -1 \le x \le 1 \\ f(x \pm 2n) & \text{otherwise where } n \in \mathbb{N}. \end{cases}$$

~5

15. Explain what Gibbs Phenomenon is, and why it happens in Fourier Series. (5pts)

When representing a piecewise function with fourier series, there will always be a "jump" at the end points where the function ends or changes. At these points, the fourier series will itself "jump" and momentarily will no longer approximate the original function. As the number of terms in series expands, so too does this discontinuity become more extreme.

4

16. What does it mean for a function to be piecewise continuous? What about piecewise smooth? (6pts)

- In order for a function to be piecewise continuous, it must be defined in terms of piecewise functions, have the limit at the potential discontinuities exist, and have the limit from the right and left exist.

i.e. $\lim_{x\to a^-} \neq \lim_{x\to a^+}$ where a exists

- For a function to be piecewise smooth, the function's derivative(s) must exist, have the derivatives limits from the left & right equal and the limit must exist. $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$

6

17. Explain the differences and similarities in the construction of a Fourier Cosine or Sine Series in comparison with the full Fourier Series. (5pts)

Any function can be represented as an infinite sum of sine and cosine terms. However if that function is odd or even, it can be represented with only sine or cosine terms respectively. This comes from the definition of the fourier series which can be defined as follows: $a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$, $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$, $b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$, and $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)$ where f is the function. If f is even or odd, then a_n or b_n will evaluate to zero. However, if f is neither even nor odd, then this trick does not work and you need the full expansion. It's also worth noting that this expansion only covers the positive terms of the expansion and not the negative.

This is all what the fun stuff

3

18. Explain how the complex Fourier series relates to the full Fourier series. (6pts)

- Just as a function can be represented an infinite sum of sine of cosine terms, so too can that function be represented as an infinite sum of complex, exponential terms, i.e. $e^{in\pi x}$. The nice thing about complex fourier series is that it only uses exponential terms as opposed to both sines and cosines. It's also worth noting that, unlike the sine and cosine series, the complex Fourier series goes from $-\infty$ to ∞, so the mathematician doesn't have to use an even or odd expansion.

ok

6

19. Set up the integrals (DO NOT evaluate) to define the coefficients for the periodic function defined by

$$f(x) = \begin{cases} x^2, & -1 < x < 1 \\ f(x+2n), & \text{otherwise} \end{cases} \qquad \text{(12pts)}$$

(a) The full Fourier series

(b) The Fourier Cosine Series (using the even half range expansion)

(c) The complex form of the Fourier series

If you make any assumptions in this, justify them.

a.) $a_0 = \frac{1}{2}\int_{-1}^{1} x^2\,dx$; $a_n = \frac{1}{1}\int_{-1}^{1} x^2 \cdot \cos\left(\frac{n\pi x}{1}\right)dx$; $b_n = \frac{1}{1}\int_{-1}^{1} x^2 \sin\left(\frac{n\pi x}{1}\right)dx = 0$

4

b.) $a_0 = \frac{1}{\cancel{2}}\int_{\cancel{-1}\,0}^{1} x^2\,dx$; $a_n = \frac{\cancel{2}}{1}\int_{\cancel{-1}\,0}^{1} x^2 \cdot \cos\left(\frac{n\pi x}{\cancel{1}}\right)dx$

3

c.) $C_n = \frac{1}{2}\int_{-1}^{1} x^2 \cdot e^{-in\pi x/1}\,dx$

4

a.) assume ~~that these integrals are on -1 to 1. Then, for $f(x \pm 2n)$, substitute $x \pm 2n$ for x to get the full expansion.~~

no...